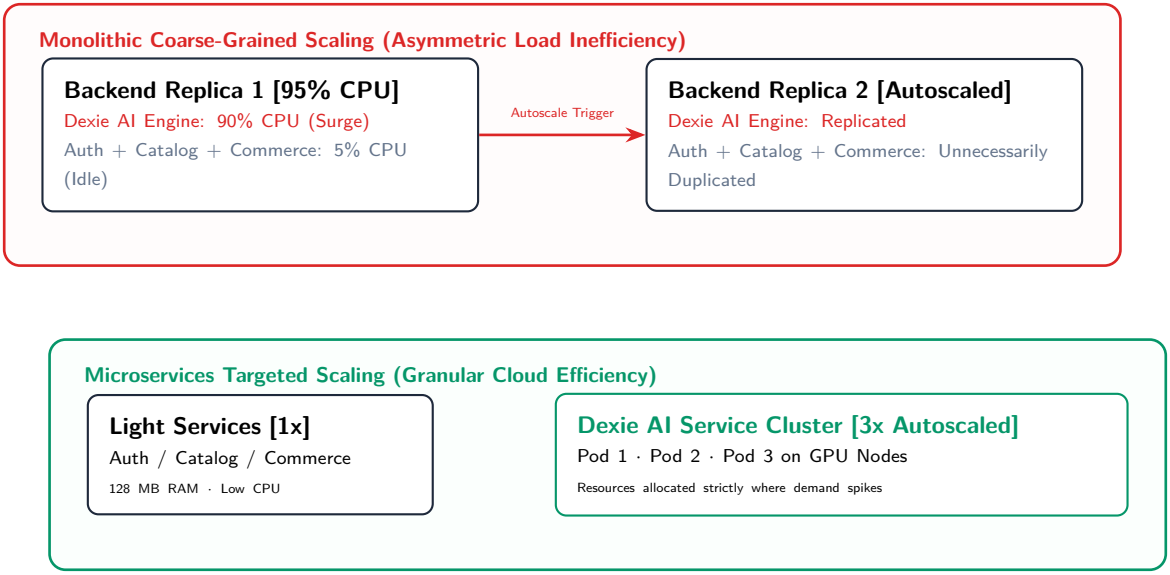
Monolithic Coarse-Grained Scaling (Asymmetric Load Inefficiency)
Backend Replica 1 [95% CPU]
Dexie AI Engine: 90% CPU (Surge)
Auth + Catalog + Commerce: 5% CPU (Idle)
Autoscale Trigger
Backend Replica 2 [Autoscaled]
Dexie AI Engine: Replicated
Auth + Catalog + Commerce: Unnecessarily Duplicated
Microservices Targeted Scaling (Granular Cloud Efficiency)
Light Services [1x]
Auth / Catalog / Commerce
128 MB RAM · Low CPU
Dexie AI Service Cluster [3x Autoscaled]
Pod 1 · Pod 2 · Pod 3 on GPU Nodes
Resources allocated strictly where demand spikes